

Task No.: 12**Date: 08-02-2026**

Title: Scanning Hotspots and Managing Wi-Fi Configurations**Aim:**

To develop an Android Application that scans hotspots and manages the Wi-Fi configurations.

Code:*MainActivity.kt*

```
package com.example.madtask12

import android.Manifest
import android.content.BroadcastReceiver
import android.content.Context
import android.content.Intent
import android.content.IntentFilter
import android.content.pm.PackageManager
import android.net.wifi.ScanResult
import android.net.wifi.WifiManager
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.content.ContextCompat

class MainActivity : AppCompatActivity() {

    private lateinit var wifiManager: WifiManager
    private lateinit var txtWifiList: TextView
    private lateinit var btnScan: Button

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)
```

```
setContentView(R.layout.activity_main)
```

```
txtWifiList = findViewById(R.id.txtWifiList)
```

```
btnScan = findViewById(R.id.btnScan)
```

```
wifiManager =
```

```
    applicationContext.getSystemService(Context.WIFI_SERVICE) as WifiManager
```

```
btnScan.setOnClickListener {
```

```
    checkPermission()
```

```
}
```

```
}
```

```
private fun checkPermission() {
```

```
    if (ContextCompat.checkSelfPermission(
```

```
        this,
```

```
        Manifest.permission.ACCESS_FINE_LOCATION
```

```
    ) != PackageManager.PERMISSION_GRANTED
```

```
    ) {
```

```
        ActivityCompat.requestPermissions(
```

```
            this,
```

```
            arrayOf(Manifest.permission.ACCESS_FINE_LOCATION),
```

```
            101
```

```
        )
```

```
    } else {
```

```
        scanWifi()
```

```
    }
```

```
}
```

```
private fun scanWifi() {
```

```
    txtWifiList.text = "Scanning Wi-Fi Networks...\n\n"
```

```
    val intentFilter =
```

```

IntentFilter(WifiManager.SCAN_RESULTS_AVAILABLE_ACTION)

registerReceiver(wifiReceiver, intentFilter)

wifiManager.startScan()
}

private val wifiReceiver = object : BroadcastReceiver() {

    override fun onReceive(context: Context, intent: Intent) {

        val results: List<ScanResult> = wifiManager.scanResults

        val builder = StringBuilder()

        builder.append("Available Wi-Fi Hotspots:\n\n")

        for (result in results) {

            val ssid = result.SSID
            val signal = result.level
            val security = getSecurityType(result)

            builder.append("SSID: $ssid\n")
            builder.append("Security: $security\n")
            builder.append("Signal Strength: $signal dBm\n")
            builder.append("-----\n")
        }

        txtWifiList.text = builder.toString()

        unregisterReceiver(this)
    }
}

private fun getSecurityType(result: ScanResult): String {

    val cap = result.capabilities

    return when {

```

```
cap.contains("WPA3") -> "WPA3"
```

```
cap.contains("WPA2") -> "WPA2"
```

```
cap.contains("WPA") -> "WPA"
```

```
cap.contains("WEP") -> "WEP"
```

```
else -> "OPEN"
```

```
}
```

```
}
```

```
override fun onRequestPermissionsResult(
```

```
    requestCode: Int,
```

```
    permissions: Array<String>,
```

```
    grantResults: IntArray
```

```
) {
```

```
    super.onRequestPermissionsResult(
```

```
        requestCode,
```

```
        permissions,
```

```
        grantResults
```

```
)
```

```
    if (requestCode == 101 &&
```

```
        grantResults.isNotEmpty() &&
```

```
        grantResults[0] == PackageManager.PERMISSION_GRANTED
```

```
) {
```

```
        scanWifi()
```

```
    } else {
```

```
        txtWifiList.text =
```

```
            "Permission Denied\nCannot Scan Wi-Fi"
```

```
    }
```

```
}
```

```
}
```

activity_main.xml

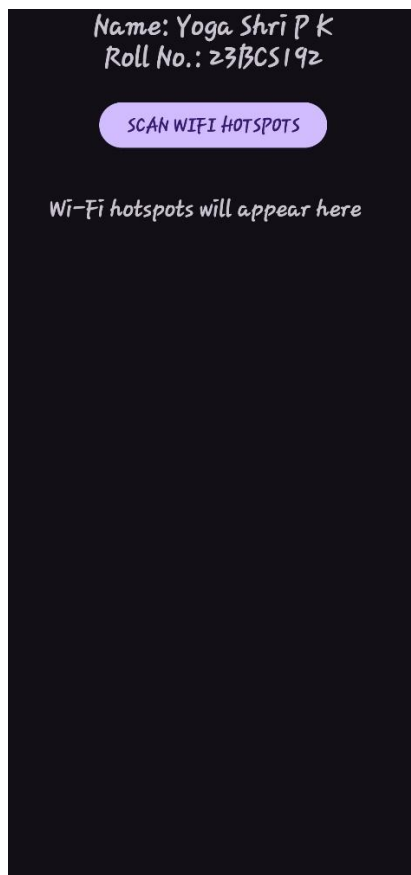
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Name: Yoga Shri P K Roll No.: 23BCS192"
        android:textSize="22sp"
        android:textStyle="bold"
        android:gravity="center"/>

    <Button
        android:id="@+id/btnScan"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:text="SCAN WIFI HOTSPOTS"
        android:layout_gravity="center"
        android:layout_marginTop="20dp"/>

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginTop="20dp">
        <TextView
            android:id="@+id/txtWifiList"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Wi-Fi hotspots will appear here"
            android:textSize="18sp"
            android:padding="16dp"/>
    </ScrollView>
</LinearLayout>
```

Output:



Result:

Thus, Android Application that scans hotspot and manages the wifi configurations is developed and executed successfully.